

Improving CelebV-HQ Face Generation through Data Curation and StyleGAN2-ADA Fine-Tuning

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GitHub: https://github.com/ahin-page/face_image_generation.git

Competition Ranking for image generative models

Abstract

This report presents our final solution for unconditional 256×256 face generation on CelebV-HQ. We compare a FLUX-based diffusion/rectified-flow baseline with a StyleGAN2-ADA based GAN and select StyleGAN2-ADA because it achieves better internal FID under the same CelebV-HQ-derived data condition. The final pipeline fine-tunes StyleGAN2-ADA from an FFHQ-256 checkpoint on a quality-filtered frame set, where removing dark and low-information samples reduces FID from 53.0515 to 33.1918 and improves IS from 3.9004 to 4.6781. The final submission uses the clean 3500king checkpoint with truncation coefficient $\psi = 1.0$ and seed 0.

1 Introduction

The CelebV-HQ [8] final challenge requires generating 1,000 unconditional 256×256 face images that match a video-derived face distribution. Unlike curated portrait datasets such as FFHQ, CelebV-HQ contains low-light frames, motion blur, black borders, imperfect crops, temporal redundancy, and wider pose/expression variation. Thus, the task is not only to generate visually appealing faces, but to match the target frame distribution.

We considered two generation directions: a FLUX-based diffusion/rectified-flow baseline [6] and a StyleGAN2-ADA [4, 5] based GAN [1]. Under the same CelebV-HQ-derived data condition, StyleGAN2-ADA achieved better internal FID, suggesting that a compact unconditional GAN with a strong face prior was better suited to this limited, noisy, narrow-domain setting.

Our final pipeline fine-tunes StyleGAN2-ADA from an FFHQ-256 checkpoint on a quality-filtered CelebV-HQ frame set. The filtering step removes severe dark and low-information frames while preserving natural video-frame variation. The final submission uses the clean 3500king checkpoint, truncation coefficient $\psi = 1.0$, and seed 0.

2 Background and Model Selection

2.1 Candidate Models Considered

Image generation has been studied through several generative modeling paradigms, including variational autoencoders, autoregressive models, normalizing flows, generative adversarial networks, and diffusion models. In this project, we focused on two practically relevant candidates for the CelebV-HQ final challenge: a FLUX-based diffusion/rectified-flow baseline and a StyleGAN2-ADA based adversarial generator. This comparison was motivated by two observations. First, diffusion-style models currently provide strong performance in large-scale image synthesis, especially in text-conditioned generation. Second, StyleGAN-style architectures have historically been highly effective for high-fidelity face generation and are particularly efficient for unconditional sampling.

2.2 GAN-based Image Generation

Generative Adversarial Networks (GANs) [1] formulate generation as a two-player minimax game between a generator G and a discriminator D . The generator maps a latent vector $z \sim p_z$ to an image $G(z)$, while the discriminator attempts to distinguish real images from generated images. The original adversarial objective is

$$\min_G \max_D \mathbb{E}_{x \sim p_{\text{data}}} [\log D(x)] + \mathbb{E}_{z \sim p_z} [\log(1 - D(G(z)))]. \quad (1)$$

At the ideal equilibrium, the generated distribution p_g matches the data distribution p_{data} . This direct distribution-matching view is well aligned with the CelebV-HQ challenge, where the goal is unconditional sampling from a target face-frame distribution.

StyleGAN [5] improves the GAN generator architecture by introducing a style-based synthesis process. Instead of feeding the input latent vector directly into the synthesis network, StyleGAN first maps

$$z \rightarrow w \quad (2)$$

through a learned mapping network. The intermediate latent code w then controls image synthesis through style modulation at multiple spatial resolutions. For face generation, this architecture is useful because different synthesis scales can represent different semantic factors: coarse layers affect global pose and face shape, middle layers affect facial parts and hairstyle, and fine layers affect texture and stochastic details. StyleGAN2 further improves this design by revising normalization and regularization to reduce artifacts and improve perceptual image quality.

For limited-data settings, however, standard GAN training can fail because the discriminator may overfit to the training set. Once the discriminator memorizes real images, it no longer provides useful gradients to the generator. StyleGAN2-ADA addresses this issue with adaptive discriminator augmentation [4]. Let $T_p(\cdot)$ denote stochastic image transformations applied with probability p . During training, both real and generated images are passed to the discriminator after augmentation:

$$D(T_p(x)), \quad D(T_p(G(z))). \quad (3)$$

The augmentation probability is adjusted automatically so that the discriminator remains useful without forcing the generator to learn the augmented distribution itself. This property is important for CelebV-HQ because the available training data are video-derived, noisy, and partially redundant.

2.3 Diffusion and Rectified-Flow Based Generation

Diffusion models [2] learn image generation through a gradual denoising process. A forward process corrupts clean images into noisy variables, and a neural network is trained to approximate the reverse process. A standard forward diffusion step can be written as

$$q(x_t | x_{t-1}) = \mathcal{N}\left(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t I\right), \quad (4)$$

where x_0 is a clean image and x_t is its noisy version at timestep t . The model then learns a reverse transition $p_\theta(x_{t-1} | x_t)$, often by predicting the added noise or an equivalent denoising target.

Rectified-flow models [7] are closely related in spirit: instead of explicitly reversing a fixed Markovian noising chain, they learn a continuous transformation or velocity field that transports a simple prior distribution to the data distribution. Modern systems such as FLUX are built on this diffusion/flow-matching family and can generate highly realistic images when trained or adapted at large scale.

However, this strength does not automatically imply superiority in the present challenge. The CelebV-HQ task is unconditional and narrow-domain. It does not require prompt following, language-image alignment, or open-world visual composition. Furthermore, adapting a large diffusion/rectified-flow model to a small and noisy target distribution can be difficult because the model must learn accurate denoising or flow behavior across many noise levels. Sampling is also iterative, whereas a GAN generates each image with a single forward pass.

2.4 Final Model Selection

We empirically compared the two candidate directions under the same CelebV-HQ-derived data condition. Table 1 summarizes the internal FID comparison.

Table 1: Internal model-selection comparison using the same CelebV-HQ-derived data.

Model	Internal FID ↓
StyleGAN2-ADA	33.94
FLUX-based baseline	53.87

The result supports our final choice of StyleGAN2-ADA. Although diffusion/rectified-flow models are powerful for large-scale image synthesis, our setting favors a compact unconditional generator with strong face-domain inductive bias and stable low-data adaptation. StyleGAN2-ADA satisfies these requirements for three reasons.

First, the adversarial objective directly encourages matching the empirical image distribution, which is aligned with the FID-based evaluation protocol. Second, the style-based generator architecture is well suited to human faces because it separates global facial structure from local texture and stochastic details across synthesis resolutions. Third, adaptive discriminator augmentation

was designed specifically to improve GAN training when the available dataset is limited, reducing discriminator memorization while preserving useful adversarial gradients.

We therefore selected StyleGAN2-ADA as the final model and focused the remaining effort on data collection, quality filtering, checkpoint selection, and deterministic inference. The final system can be interpreted as a transfer-learning pipeline: FFHQ-256 provides a strong prior over realistic face geometry and texture, while CelebV-HQ fine-tuning adapts the generator to the target video-frame distribution.

3 Implementation Details and Methodology

3.1 Dataset and Data Collection

We used the official CelebV-HQ dataset released by the dataset authors. The dataset was downloaded from the public CelebV-HQ repository: <https://github.com/CelebV-HQ/CelebV-HQ>. No additional face images were scraped from the internet, and no real images from external image-generation services or unrelated datasets were used for the final submission.

Since CelebV-HQ is distributed as a video-based face dataset, we first constructed an image-frame dataset for unconditional image generation. Frames were sampled from the downloaded CelebV-HQ videos and converted into face-centered 256×256 images.

The video-derived nature of CelebV-HQ made the raw frame distribution substantially different from curated face datasets such as FFHQ. The raw frames contained temporal redundancy between nearby video frames, low-light scenes, motion blur, black image borders, imperfect crops, and frames where the face was only weakly visible. These artifacts are not merely cosmetic issues: because a generative model learns the empirical training distribution, frequent low-information frames can become valid modes that the model attempts to reproduce. Indeed, direct StyleGAN2-ADA training on the unfiltered frame set produced dark and weakly structured generated images.

3.2 Quality-Aware Dataset Curation

To reduce pathological training modes while preserving the CelebV-HQ video-frame character, we applied deterministic quality filtering based on simple image statistics. For each sampled frame, we computed mean luminance, center luminance, contrast, dark-pixel fraction, bright-pixel fraction, and edge strength. The filtering criteria are summarized in Table 2.

Table 2: Quality filtering thresholds used to construct the final training set.

Metric	Threshold
Mean luminance	$0.18 \leq \mu \leq 0.78$
Center luminance	$\mu_{\text{center}} \geq 0.16$
Contrast	$\sigma \geq 0.08$
Dark-pixel fraction	≤ 0.40
Bright-pixel fraction	≤ 0.08
Edge strength	≥ 0.012

Out of the 200,000 raw frames, 133,064 frames passed the quality filter. We then selected 100,000 images using seed 0 to form the final clean training subset. During StyleGAN2-ADA training, horizontal mirroring was enabled, resulting in an effective training set size of 200,000 images.

Figure 1 visualizes the effect of the filtering procedure. The mean-luminance distribution shows that the low-luminance tail was removed, while the dark-pixel-fraction distribution shows that frames with excessive dark regions were suppressed. This directly addresses the failure mode observed in the unfiltered GAN model, where the generator learned to produce dark and low-information faces.

The filtering strategy was deliberately moderate. Its objective was not to transform CelebV-HQ into a clean portrait dataset, but to remove the most severe low-information samples that caused degenerate generator behavior. Overly aggressive filtering could improve visual cleanliness while shifting the training distribution away from the target CelebV-HQ distribution. Therefore, the final thresholds were chosen to remove the pathological tail while preserving natural variation in lighting, pose, and facial appearance.

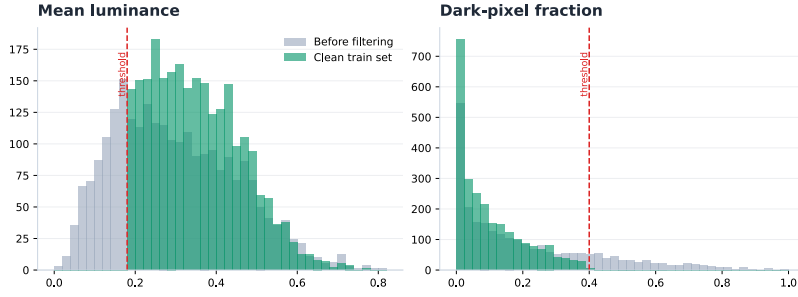


Figure 1: Effect of quality filtering on the raw CelebV-HQ frame distribution. The left histogram compares mean luminance before and after filtering, and the right histogram compares dark-pixel fraction. The dashed red lines indicate the thresholds used for filtering.

3.3 Model Training and Fine-Tuning

We trained two candidate image-generation pipelines: a FLUX-based diffusion/rectified-flow baseline and a StyleGAN2-ADA based adversarial generator. The FLUX baseline was adapted to the CelebV-HQ frame domain using LoRA fine-tuning. LoRA updates only a low-rank set of trainable adaptation parameters while keeping most of the pretrained model fixed, making it a practical choice for adapting a large pretrained diffusion/flow model under limited compute and data [3]. However, despite this parameter-efficient adaptation, the FLUX-based baseline achieved a weaker internal FID than the GAN-based model under the same CelebV-HQ-derived data condition.

Our final model was StyleGAN2-ADA fine-tuned from an FFHQ-256 pretrained checkpoint. This initialization provides a strong prior over realistic face geometry, identity structure, and local facial texture. The role of fine-tuning is then to adapt the pretrained face manifold to the CelebV-HQ video-frame distribution, which contains less controlled illumination, broader pose variation, motion blur, and imperfect frame-level artifacts.

StyleGAN2-ADA was trained on the curated `data/faces_200k_clean` subset at 256×256 resolution with adaptive discriminator augmentation and horizontal mirroring enabled. We submitted and evaluated multiple training checkpoints; among them, the clean 3500king checkpoint produced the best leaderboard score profile. Compared with the clean 2500king checkpoint, the 3500king model improved FID, IS, and KID, while incurring only a small decrease in TopPR. The final submitted images were generated from this best-performing StyleGAN2-ADA checkpoint using seed 0. Detailed inference settings, checkpoint hash, and the exact generation command are provided in the appendix for reproducibility.

4 Results

4.1 Model Selection Result

We first compared the two candidate generation directions under the same CelebV-HQ-derived data condition. The StyleGAN2-ADA model achieved an internal FID of 33.94, while the FLUX-based LoRA baseline achieved an internal FID of 53.87. This result supported the choice of StyleGAN2-ADA as the final submission model. The following experiments therefore focus on StyleGAN2-ADA checkpoints and the effect of dataset curation.

4.2 Checkpoint Comparison

Table 3 summarizes the leaderboard metrics for the evaluated StyleGAN2-ADA candidates. We report FID, IS, KID, and TopPR because the models showed different behavior across fidelity and coverage-oriented metrics.

Table 3: Leaderboard metrics for evaluated StyleGAN2-ADA checkpoints.

Run	FID ↓	IS ↑	KID ↓	TopPR ↑
Before cleaning	53.0515	3.9004	0.0039	0.8377
Clean 2500king	33.9015	4.6353	0.0046	0.8198
Clean 3500king	33.1918	4.6781	0.0044	0.8167

The clean 3500king checkpoint achieved the best FID, IS, and KID among the evaluated StyleGAN2-ADA candidates. Compared with the clean 2500king checkpoint, FID decreased from 33.9015 to 33.1918, IS increased from 4.6353 to 4.6781,

and KID decreased from 0.0046 to 0.0044. TopPR decreased slightly from 0.8198 to 0.8167. Among our StyleGAN2-ADA submissions, the clean 3500king checkpoint achieved the best FID, IS, and KID, with only a small decrease in TopPR, and was therefore used for the final submission.

4.3 Qualitative Result

Figure 2 shows generated examples from the final clean 3500king checkpoint.

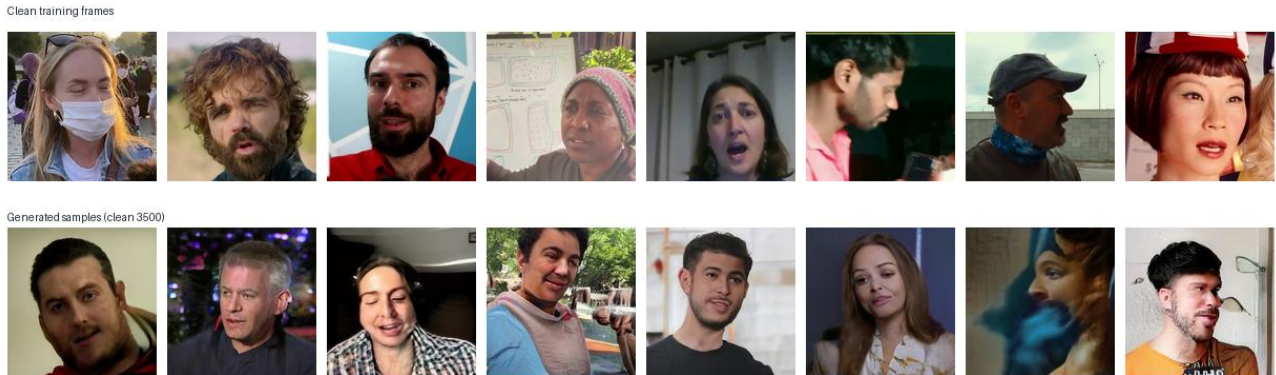


Figure 2: Generated samples from the final clean 3500king StyleGAN2-ADA checkpoint.

The final model produces centered and recognizable faces, though the samples remain noticeably imperfect, exhibiting residual blur and inconsistent illumination. Incorporating additional clean face datasets would likely improve visual quality, but this task evaluates distributional similarity against CelebV-HQ specifically — the quality gap largely reflects the noisy, video-derived nature of the target distribution itself.

5 Analysis

5.1 Failure Cases and Lessons

Table 4 summarizes the main failure cases observed during development.

Table 4: Failure cases and corresponding lessons.

Failure case	Cause	Lesson
Dark-frame generation	Raw video frames contained near-black and low-information samples.	Dataset quality directly affects the learned generator distribution.
Over-cleaning risk	Removing all imperfect frames shifts the data toward FFHQ-like portraits.	Filtering should remove pathological tails, not natural CelebV-HQ artifacts.
TopPR decrease	Longer training and larger truncation improved some fidelity metrics but reduced coverage.	Checkpoint and inference choices must balance fidelity and recall-like metrics.
FLUX baseline gap	LoRA adaptation of a large diffusion/flow model was less effective in this narrow, noisy, unconditional setting.	A compact domain-specific GAN was better matched to the challenge constraints.
Reproducibility issue	Early same-seed runs were affected by backend non-determinism.	Final inference must fix seeds and deterministic backend settings.

Overall, the final score was determined by distribution control rather than model scale alone. The most important decisions were selecting a low-data-robust GAN, filtering pathological video frames, choosing the 3500king checkpoint, and using $\psi = 1.0$ for final inference.

5.2 Truncation Coefficient and Distribution Matching

We analyzed the effect of the StyleGAN2-ADA truncation coefficient ψ using the final clean 3500king checkpoint. The truncation coefficient is an inference-time sampling parameter that controls how strongly latent codes are pulled toward the average

latent code. Conceptually, the truncated latent code can be written as

$$w' = \bar{w} + \psi(w - \bar{w}), \quad (5)$$

where w is the sampled latent code, $\bar{w}$ is the average latent code, and w' is the latent code used for generation. Smaller ψ values produce more conservative samples by emphasizing average facial characteristics, while larger ψ values preserve more latent-space variation.

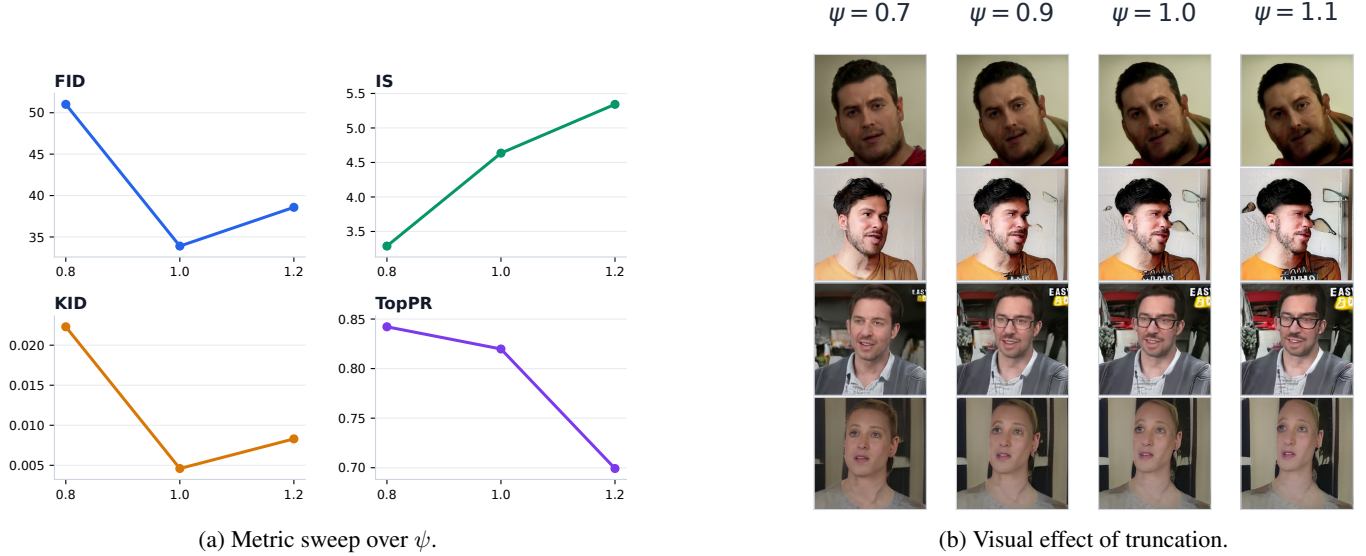


Figure 3: Effect of the truncation coefficient on final generation. The metric sweep shows that $\psi = 1.0$ minimizes FID and KID, while larger ψ improves IS but reduces TopPR. The visual comparison shows that changing ψ modifies facial attributes and local image details even when the underlying seed is fixed.

Interestingly, $\psi = 0.8$ produced samples that were visually clean and subjectively appealing, but it did not achieve the best distributional metrics. At $\psi = 0.8$, TopPR was highest, but FID, IS, and KID degraded significantly. This suggests that overly conservative truncation collapsed samples toward a narrow central latent region, producing attractive but less distribution-matched images. At $\psi = 1.2$, IS increased, but TopPR dropped sharply, indicating that broader latent sampling produced more confident or varied samples while reducing coverage of the reliable CelebV-HQ distribution. We selected $\psi = 1.0$ because it achieved the best FID and KID while maintaining relatively high TopPR, making it the most appropriate setting for distribution matching rather than subjective visual preference alone.

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